

| QATCH openQCM Real-Time FW Change Control Document |            |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------|------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Release Version                                    | Date       | Author  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>v2.7b1</b>                                      | 05/06/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Initial beta version for the v2.7x branch</li> <li>- Updated build information and updater client</li> <li>- Temperature instability during cooldown</li> <li>- Automatic cartridge lock with manual override</li> <li>- Support lid switch for dynamic position detection</li> <li>- Various temperature bugs fixed with new correction</li> </ul> |
| <b>v2.7r1</b>                                      | 05/06/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Updated build information</li> <li>- Same change log as v2.7b1</li> </ul>                                                                                                                                                                                                                                                                           |
| <b>v2.7b2</b>                                      | 05/06/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Updated build information</li> <li>- No functional changes from v2.7b1</li> </ul>                                                                                                                                                                                                                                                                   |
| <b>v2.7r2</b>                                      | 05/06/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Updated build information</li> <li>- No functional changes from v2.7r1</li> </ul>                                                                                                                                                                                                                                                                   |
| <b>v2.7b3</b>                                      | 05/18/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Updated build information</li> <li>- No functional changes from v2.7b2</li> </ul>                                                                                                                                                                                                                                                                   |
| <b>v2.7r3</b>                                      | 05/18/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Updated build information</li> <li>- No functional changes from v2.7r2</li> </ul>                                                                                                                                                                                                                                                                   |
| <b>v2.7b4</b>                                      | 05/22/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Updated build information</li> <li>- No functional changes from v2.7b3</li> </ul>                                                                                                                                                                                                                                                                   |

## FW Changes

|               |            |         |                                                                                                                                                                                                                                                                                                                               |
|---------------|------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>v2.7r4</b> | 05/22/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Updated build information</li> <li>- No functional changes from v2.7r3</li> </ul>                                                                                                                    |
| <b>v2.7b5</b> | 07/09/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Updated build information</li> <li>- No functional changes from v2.7b4</li> </ul>                                                                                                                    |
| <b>v2.7b6</b> | 07/15/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Updated build information</li> <li>- No functional changes from v2.7b5</li> </ul>                                                                                                                    |
| <b>v2.7r6</b> | 07/15/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Updated build information</li> <li>- No functional changes from v2.7r5</li> </ul>                                                                                                                    |
| <b>v2.7b7</b> | 07/21/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Updated build information</li> <li>- No functional changes from v2.7b6</li> </ul>                                                                                                                    |
| <b>v2.7r7</b> | 07/21/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Updated build information</li> <li>- No functional changes from v2.7r6</li> </ul>                                                                                                                    |
| <b>v2.7b8</b> | 09/08/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Updated build information</li> <li>- Improve user interactions with bug fixes</li> <li>- Fix #432: Lid movements out-of-sync at startup</li> <li>- Fix #439: Clear MSGBOX on LID movement</li> </ul> |
| <b>v2.7r8</b> | 09/08/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Updated build information</li> <li>- Improve user interactions with bug fixes</li> <li>- Fix #432: Lid movements out-of-sync at startup</li> <li>- Fix #439: Clear MSGBOX on LID movement</li> </ul> |
|               |            |         |                                                                                                                                                                                                                                                                                                                               |